

Mock Space

Sample Paper - Biology

Answer all questions. 45 minutes. 30 marks.

1 Define the term osmosis. **[2]**

2 A student heats an enzyme solution to 65 °C and the reaction stops.
Explain, in terms of protein structure, why the reaction stops. **[4]**

3 State two variables that must be controlled in this investigation. **[2]**

4 (a) Describe how you would test a leaf for the presence of starch. **[4]**

4 (b) Explain why the leaf must be boiled in ethanol during this test. **[3]**

5 The rate of photosynthesis increases with light intensity, then plateaus.
Suggest what limits the rate at the plateau. **[3]**

6 Compare aerobic and anaerobic respiration in muscle cells. **[6]**

7 Evaluate the use of antibiotics in intensive farming. **[6]**
